

Gradient Bandit Algorithms

- Use parameterized PMF over actions
- PMF is a function of an action preference function $H_t(a)$, $a \in A$.
 - A certain evaluation of an action

PMF is parametrized by $H_t(a)$.
- Update evaluation based on obtained rewards
 - This updates the PMF
- An instance of stochastic gradient descent

Gradient Bandit Algorithms

- Use parameterized PMF over actions

$\pi_t(a) = P[A_t = a]$, $a \in \mathcal{A}$, is the PMF at step t .

Write $E[R_t]$ in terms of $\pi_t(a)$.

$$E[R_t] = \sum_a E[R_t | A_t = a] \pi_t(a) = \sum_a \phi^*(a) \pi_t(a)$$

Gradient Bandit Algorithms

- Use parameterized PMF over actions

$\pi_t(a) = P[A_t = a]$, $a \in \mathcal{A}$, is the PMF at step t .

We want to maximize
 $E[R_t]$ over the space of policy PMFs.

→ Parameterize the policy PMF

→ Optimize over the space of parameters.

Parameters: $\Pi_t(a)$, $a \in A$.

PMF (softmax)

$$\Pi_t(a) = \frac{e^{\Pi_t(a)}}{\sum_{b \in A} e^{\Pi_t(b)}}, \quad a \in A$$

Goal is to maximize

$E[R_t]$ over the

parameters $\Pi_t(a)$, $a \in A$.

We know how to maximize $f(\theta)$ over the parameter vector $\theta_{1 \times 1}$

First order necessary condition:

$$\nabla f(\theta) = 0 \text{ at } \theta^*$$

$$\text{Let } g(\theta) = \nabla f(\theta)$$

We want to solve $g(\theta) = 0$.

Start with a guess θ_0 . In the neighborhood of θ_0 , the following is a good approximation:

$$g(\theta) \approx g(\theta_0) + (\theta - \theta_0)^T \nabla g(\theta_0) \quad [\text{Scalar parameter}]$$

$$g(\theta) \approx g(\theta_0) + \overbrace{\nabla g(\theta_0)}^{\text{Jacobian}} (\theta - \theta_0)_{1 \times 1} \quad \left\{ \text{Vector } \theta \right\}$$

How do we get the above?

What do we do next?

Setting $g(\theta) = 0$, gives us:

$$0 = g(\theta_0) + \nabla g(\theta_0) (\theta - \theta_0)$$

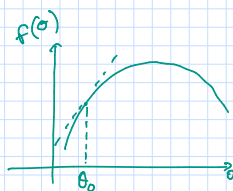
$$\nabla g(\theta_0) \theta = -g(\theta_0) + \nabla g(\theta_0) \theta_0$$

$$\theta_1 = -\nabla g(\theta_0)^{-1} g(\theta_0) + \theta_0$$

$$= \theta_0 - \nabla g(\theta_0)^{-1} g(\theta_0)$$

$$= \theta_0 - [\nabla^2 f(\theta_0)]^{-1} g(\theta_0)$$

$$= \theta_0 - [\nabla^2 f(\theta_0)]^{-1} \nabla f(\theta_0)$$



GRADIENT ASCENT

$$\theta_1 = \theta_0 + \alpha \nabla f(\theta_0)$$

$$\theta_2 = \theta_1 + \alpha \nabla f(\theta_1)$$

$\vdots$

Back to our problem

$$\text{maximize } E[R_t] \\ (H_t(1), H_t(2), \dots, H_t(k))$$

We want to solve

$$\nabla_{\bar{H}_t(\alpha)} E[R_t] = 0$$

Just do:

$$\begin{aligned} \bar{H}_{t,1}(\alpha) &= \bar{H}_{t,0}(\alpha) + \alpha \nabla E[R_t] \\ &\vdots \\ &\vdots \end{aligned}$$

Is there a problem?

Use Noisy Estimates of $\nabla_{\vec{\theta}_t} E[R_t]$

Gradient Ascent $\rightarrow$ Stochastic Gradient Descent

Your step updates are "noisy".

Let

$$\tilde{\nabla}_{\vec{\theta}_t} E[R_t] \approx \nabla_{\vec{\theta}_t} E[R_t]$$

It is desirable that the noisy estimates satisfy the property:

$$E[\tilde{\nabla}_{\vec{\theta}_t} E[R_t]] = \nabla_{\vec{\theta}_t} E[R_t]$$

Turns out that the following update step satisfies the above property:

$$\vec{\theta}_{t+1}(a) = \vec{\theta}_t(a) + \alpha (R_t - \bar{R}_t) (\mathbb{I}_{\{\vec{\theta}_t(a) = a\}} - \pi_t(a))$$

for every a in $\mathcal{A}$.

Gradient Bandit Algorithms

[Building Intuition]

- The update step

$$H_{t+1}(a) = H_t(a) + \alpha(R_t - \bar{R}_t)(\mathbb{I}_{\{A_t=a\}} - \pi_t(a)), \forall a.$$

If $A_t = a$:

$$H_{t+1}(a) = H_t(a) + \alpha(R_t - \bar{R}_t)(1 - \pi_t(a))$$

If $A_t \neq a$:

$$H_{t+1}(a) = H_t(a) + \alpha(R_t - \bar{R}_t)(-\pi_t(a))$$

Gradient Bandit Algorithms

- PMF is a function of a certain evaluation of an action
 - Why not choose Q values as the action preference function?

Gradient Bandit Algorithms

- An instance of stochastic gradient ~~descent~~
ascent

We want to show that for every action a ,

$$\frac{\partial E[R_t]}{\partial \pi_t(a)} = E[(R_t - \bar{R}_t)(I_{\{A_t=a\}} - \pi_t(a))]$$

Begin by showing that

$$\frac{\partial E[R_t]}{\partial \pi_t(a)} = \frac{\partial}{\partial \pi_t(a)} \left(\sum_{a'} \pi_t(a') (\phi^*(a') - B_t) \right),$$

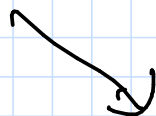
where B_t is a baseline constant w.r.t. $\pi_t(a)$.

Consider

$$\begin{aligned} \frac{\partial}{\partial \pi_t(a)} \sum_{a'} \pi_t(a') B_t \\ = B_t \frac{\partial}{\partial \pi_t(a)} \sum_{a'} \pi_t(a') \\ = 0 \end{aligned}$$

$$\frac{\partial E[R_t]}{\partial \pi_t(a)} = \frac{\partial}{\partial \pi_t(a)} \sum_{b \in A} \phi^*(b) \pi_t(b)$$

$$= \sum_{b \in A} \phi^*(b) \frac{\partial \pi_t(b)}{\partial \pi_t(a)}$$



Hence proved.

Rewrite

$\frac{\partial}{\partial H_t(a)} \left(\sum_{a'} \pi_t(a') (\phi^*(a') - B_t) \right)$ as an expected value.

$$\begin{aligned}
 & \sum_{a'} \frac{\partial \pi_t(a')}{\partial H_t(a)} (\phi^*(a') - B_t) \\
 &= \sum_{a'} \left[\frac{\partial \pi_t(a')}{\partial H_t(a)} (\phi^*(a') - B_t) \right] \frac{\pi_t(a')}{\pi_t(a')} \rightarrow g(a') \\
 &= \mathbb{E}_{A_t} \left[(\phi^*(A_t) - B_t) \frac{1}{\pi_t(A_t)} \frac{\partial \pi_t(A_t)}{\partial H_t(a)} \right] \\
 & \quad \text{Replace by } \bar{R}_t
 \end{aligned}
 \left| \begin{aligned}
 & \sum_{a'} g(a') \underbrace{P[A_t = a']}_{\pi_t(a')} \\
 & \triangleq \mathbb{E}[g(A_t)]
 \end{aligned} \right.$$

$$\begin{aligned}
 & \sum_{a'} \frac{\partial \pi_t(a')}{\partial H_t(a)} (\phi^*(a') - B_t) \\
 &= \sum_{a'} \frac{\partial \pi_t(a')}{\partial H_t(a)} (\mathbb{E}[R_t | A_t = a'] - B_t) \frac{\pi_t(a')}{\pi_t(a')} \\
 &= \sum_{a'} \left(\frac{1}{\pi_t(a')} \frac{\partial \pi_t(a')}{\partial H_t(a)} (\mathbb{E}[R_t | A_t = a'] - B_t) \right) \pi_t(a') \\
 &= \mathbb{E}_{A_t} \left[\frac{1}{\pi_t(A_t)} \frac{\partial \pi_t(A_t)}{\partial H_t(a)} (R_t - B_t) \right]
 \end{aligned}$$

$$\mathbb{E}[X | Y=y]$$

$$\sum \mathbb{E}[X | Y=y] P[Y=y] = \mathbb{E}[X]$$

We should obtain

$$\sum_{b \in A} (\phi^*(b) - \beta_t) \frac{\partial \pi_t(b)}{\partial H_t(a)} = E_{A_t} \left[(\phi^*(A_t) - \beta_t) \frac{\partial \pi_t(A_t)}{\partial H_t(a)} \frac{1}{\pi_t(A_t)} \right]$$

Next, to complete the proof, show that:

$$\frac{\partial \pi_t(A_t)}{\partial H_t(a)} \frac{1}{\pi_t(A_t)} = I_{\{A_t = a\}} - \pi_t(A_t).$$

Markov Decision Processes

Capturing The Stochastic (Markov) Environment

Recall The Optimization Problem

- Multi-stage optimization in a stochastic environment
- Sequential decision making
 - Time (*Discrete time*)
 - On what basis? *State S_t at time t .*
 - By whom? *Agent*
 - In what setting? *Env.*
 - What guides the agent to get better at achieving its goal?
 $R_{t+1}, R_{t+2}, \dots$
 - Who sets the goal!?

State, Action, Reward, Next State, Next Action...

- What all must influence the reward and the next state?

Observe S_t and take action A_t at time t as a function of S_t .

Obtain reward R_{t+1} & the env transitions to state S_{t+1} .

Agent - Environment Interaction

See the book for a
nice illustration.

Capturing The Stochastic (Markov) Environment

- The **Markov** Property
 - The future is entirely determined by the current state and your action
 - Given the current, no amount of information from the past is beneficial